North South University

Department of Electrical & Computer Engineering

PROJECT PROPOSAL

Course Code	CSE311L
Course Title	Database Systems Lab
Faculty Name	Ms. Tanzilah Noor Shabnam [TnS1]
Lab Instructor	Md. Amanullah Shah
Section	05
Group Number	02
Date of Submission	2026-03-13
Group Members Name & ID	Abu Md Ashraful Rahat: 2232755642 Subaita Maryam: 2222743642 Md. Al-Amin: 2131585642 Rouzatul Romman: 2031624642

Project Proposal — Submission Form

1. Project Title & One-Line Description

Title: Smart Traffic Complaint and Reporting System for Bangladesh

One-Line Description: A web-based system that allows citizens to report traffic-related problems such as illegal parking, broken signals, road blockages, and accidents, while enabling traffic authorities to manage and monitor these complaints through a centralized database platform.

2. Problem Statement

The proposed **Smart Traffic Complaint and Reporting System** is a centralized web platform designed to digitize traffic management in Bangladesh. It allows citizens to formally report issues like **illegal parking and rule violations**, moving away from inefficient social media complaints. By using a **centralized database**, authorities can efficiently track, review, and resolve documented traffic problems in real time.

3. Core Features / Scope

I. **User Registration** – Citizens can create accounts using personal details.
II. **User Login** – Registered users can log in securely.
III. **Submit Traffic Complaint** – Users can report traffic issues with location and description.
IV. **Complaint Categorization** – Traffic issues are organized by category.
V. **Complaint Tracking** – Users can track complaint status.
VI. **View Complaint History** – Users can view previously submitted complaints.
VII. **Admin Dashboard** – Administrators can monitor complaints in a central panel.
VIII. **Complaint Status Management** – Admin can update status (Pending, In Progress, Solved).
IX. **Complaint Filtering** – Admin can filter complaints by category, location, or date.
X. **Complaint Deletion** – Admin can remove invalid or duplicate complaints.
XI. **Notification System** – Users receive updates when complaint status changes.
XII. **Traffic Issue Analytics** – The system generates summaries of traffic problems.

The technical implementation is based on the following tools:

I. HTML – Used to structure web pages.
II. CSS – Used for visual styling and layout.
III. PHP – Used for server-side processing and application logic.
IV. MySQL / SQL – Used for storing and managing system data.

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Database Entities and Attributes:

Users:

1. user_id
2. full_name
3. email
4. phone_number
5. password
6. address
7. city

8. registration_date
9. last_login
10. account_status
11. profile_photo
12. user_role

Admin:

1. admin_id
2. admin_name
3. email
4. phone
5. password
6. department
7. designation
8. joining_date
9. last_login
10. account_status
11. access_level
12. office_location

Complaint:

1. complaint_id
2. user_id
3. category_id
4. location_id
5. description
6. priority_level
7. submission_date
8. last_updated
9. status_id
10. admin_id
11. evidence_file
12. resolution_note

Complaint_Categories:

1. category_id
2. category_name
3. description
4. created_date
5. updated_date
6. status
7. priority_level
8. department
9. category_code
10. category_type
11. created_by
12. remarks

Complaint_Status:

1. status_id
2. status_name
3. description
4. created_date
5. updated_date
6. status_level
7. is_active
8. status_code
9. handled_by
10. remarks
11. last_modified
12. status_type

Location:

1. location_id
2. road_name
3. area
4. district
5. city
6. postal_code
7. landmark
8. latitude
9. longitude
10. traffic_zone
11. created_date
12. updated_date

attachment:

1. attachment_id
2. complaint_id
3. file_name
4. file_type
5. file_size
6. upload_date
7. uploaded_by
8. file_path
9. file_status
10. verification_status
11. remarks
12. last_updated

Notification:

1. notification_id
2. user_id
3. complaint_id
4. message
5. notification_type
6. sent_date
7. read_status
8. delivery_status
9. priority
10. channel

11. created_by
12. remarks

Complaint_History:

1. history_id
2. complaint_id
3. old_status
4. new_status
5. changed_by
6. change_date
7. remarks
8. update_source
9. priority_change
10. assigned_admin
11. resolution_time
12. status_note

Traffic_Departments:

1. department_id
2. department_name
3. city
4. office_address
5. phone_number
6. email
7. head_of_department
8. staff_count
9. created_date
10. updated_date
11. department_code
12. status

Report:

1. report_id
2. report_title
3. report_type
4. generated_by
5. generated_date
6. total_complaints
7. resolved_complaints
8. pending_complaints
9. location_filter
10. category_filter
11. report_file
12. remarks

Login_Logs:

1. log_id
2. user_id
3. login_time
4. logout_time
5. ip_address
6. device_type
7. browser
8. location

9. login_status
10. failed_attempts
11. security_flag
12. remarks

Entity Relationships

I. Users (1) — (Many) Complaints: A user can submit multiple complaints.
II. Complaints (Many) — (1) Complaint_Categories: Each complaint belongs to one category.
III. Complaints (Many) — (1) Locations: Multiple complaints may occur at the same location.
IV. Admins (1) — (Many) Complaints: An admin may manage many complaints.
V. Complaints (1) — (Many) Attachments: A complaint may include several files.
VI. Complaints (1) — (Many) Complaint_History: Each complaint can have multiple updates.
VII. Users (1) — (Many) Notifications: Users may receive multiple notifications.

System Page:

I. Home Page
II. User Registration Page
III. Login Page
IV. User Dashboard
V. Submit Complaint Page
VI. Complaint History Page
VII. Complaint Details Page
VIII. Admin Dashboard
IX. Complaint Management Page
X. Reports and Analytics Page

System Architecture:

The proposed system follows a three-layer architecture consisting of the presentation layer (HTML and CSS), the application layer (PHP), and the database layer (MySQL). The user interacts with the system through web pages designed with HTML and CSS. PHP processes the user requests and communicates with the MySQL database to store, retrieve, and update information related to users and complaints.

4. Team Roles

Student Name	Student ID	Role
Abu Md Ashraful Rahat	2232755642	HTML, CSS, Merge and Authentications
Subaita Maryam	2222743642	MySql
Md. Al-Amin	2131585642	PHP
Rouzatul Romman	2031624642	HTML, CSS